Compressible flow

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \underline{v}) = 0 \qquad \text{[COM v1]}$$

$$\partial_t \rho + \partial_j(\rho v_j) = 0$$

equivalently

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \underline{v} = 0 \qquad \text{[COM v2]}$$

$$\partial_t \rho + v_j \partial_j \rho + \rho \partial_j v_j = 0$$

$$\frac{\partial(\rho \underline{v})}{\partial t} + \nabla \cdot (\rho \underline{v}\,\underline{v}) = \rho \underline{F} + \nabla \cdot \underline{\underline{T}} \qquad \text{[COLM v1]}$$

$$\partial_t(\rho v_i) + \partial_j(\rho v_j v_i) = \rho F_i + \partial_j T_{ji}$$

equivalently (using COM to simplify)

$$\rho \frac{D\underline{v}}{Dt} = \rho \underline{F} + \nabla \cdot \underline{\underline{T}} \qquad \text{[COLM v2]}$$

$$\rho \left(\partial_t v_i + v_j \partial_j v_i\right) = \rho F_i + \partial_j T_{ji}$$

Incompressible flow

$$\nabla \cdot \underline{v} = 0 \qquad \text{[COM]}$$

$$\partial_j v_j = 0$$

$$\rho_0 \left[\frac{\partial \underline{v}}{\partial t} + \nabla \cdot (\underline{v}\,\underline{v})\right] = \rho_0 \underline{F} + \nabla \cdot \underline{\underline{T}} \qquad \text{[COLM v1]}$$

$$\rho_0 \left[\partial_t v_i + \partial_j(v_j v_i)\right] = \rho_0 F_i + \partial_j T_{ji}$$

equivalently (using COM to simplify)

$$\rho_0 \frac{D\underline{v}}{Dt} = \rho_0 \underline{F} + \nabla \cdot \underline{\underline{T}} \qquad \text{[COLM v2]}$$

$$\rho_0 \left(\partial_t v_i + v_j \partial_j v_i\right) = \rho_0 F_i + \partial_j T_{ji}$$

Don't be deceived because the last eq. looks just like its left counterpart. Here $\rho_0 = \mathit{const.}$ (we're not solving for it), on the left $\rho = \rho(x, y, z, t)$ (we're solving for it).

The stress tensor of a ***Newtonian fluid*** (terms vanish for *incompressible* flows) in Cartesian coords $(x_1, x_2, x_3) = (x, y, z)$ with velocity $\underline{v} = \big(v_1(x,y,z), v_2(x,y,z), v_3(x,y,z)\big)$:

$$\underline{\underline{T}} = \begin{pmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{pmatrix} + \begin{pmatrix} 2\mu\frac{\partial v_1}{\partial x} - \frac{2}{3}\mu\nabla\cdot\underline{v} & \mu\left(\frac{\partial v_2}{\partial x} + \frac{\partial v_1}{\partial y}\right) & \mu\left(\frac{\partial v_3}{\partial x} + \frac{\partial v_1}{\partial z}\right) \\ \mu\left(\frac{\partial v_1}{\partial y} + \frac{\partial v_2}{\partial x}\right) & 2\mu\frac{\partial v_2}{\partial y} - \frac{2}{3}\mu\nabla\cdot\underline{v} & \mu\left(\frac{\partial v_3}{\partial y} + \frac{\partial v_2}{\partial z}\right) \\ \mu\left(\frac{\partial v_1}{\partial z} + \frac{\partial v_3}{\partial x}\right) & \mu\left(\frac{\partial v_2}{\partial z} + \frac{\partial v_3}{\partial y}\right) & 2\mu\frac{\partial v_3}{\partial z} - \frac{2}{3}\mu\nabla\cdot\underline{v} \end{pmatrix}$$

$$T_{ij} = -p\delta_{ij} - \tfrac{2}{3}\mu\partial_k v_k\delta_{ij} + 2\mu\partial_{(i}v_{j)}, \qquad \text{[\underline{total} (Cauchy) stress tensor]}$$

The Newtonian *viscous* stress tensor is $\tau_{ij} = -\frac{2}{3}\mu\partial_k v_k\delta_{ij} + 2\mu\partial_{(i}v_{j)}$.
Note: $\underline{\underline{\tau}}$ has *nonzero* diagonal components, however, because of our *definition* $p = -\frac{1}{3}\operatorname{tr}\underline{\underline{T}} = -\frac{1}{3}T_{ii}$ we must have $\operatorname{tr}\underline{\underline{\tau}} = \tau_{ii} = \tau_{11} + \tau_{22} + \tau_{33} = 0$ for *both compressible and incompressible* flows, requiring the use of Stokes' hypothesis (that is, $\lambda = -2\mu/3$).
Note: ρ and μ do *not* have to be constant here, and this fluid could be compressible, in which case an *equation of state* is needed to relate p and ρ (assuming isothermal conditions).

For constant properties, $\rho = \rho_0$ and $\mu = \mu_0$, using the above $\underline{\underline{T}}$ in COLM v2, we obtain the ***Navier–Stokes equations for an incompressible flow of a Newtonian fluid***:

$$\nabla\cdot\underline{v} = 0, \qquad \rho_0\frac{D\underline{v}}{Dt} = \rho_0\underline{F} - \nabla p + \mu_0\nabla^2\underline{v} \qquad [\rho_0, \mu_0 = const.] \qquad \text{[iNS]}$$

$$\partial_j v_j = 0, \qquad \rho_0\left(\partial_t v_i + v_j\partial_j v_i\right) = \rho_0 F_i - \partial_i p + \mu_0\partial_j\partial_j v_i$$